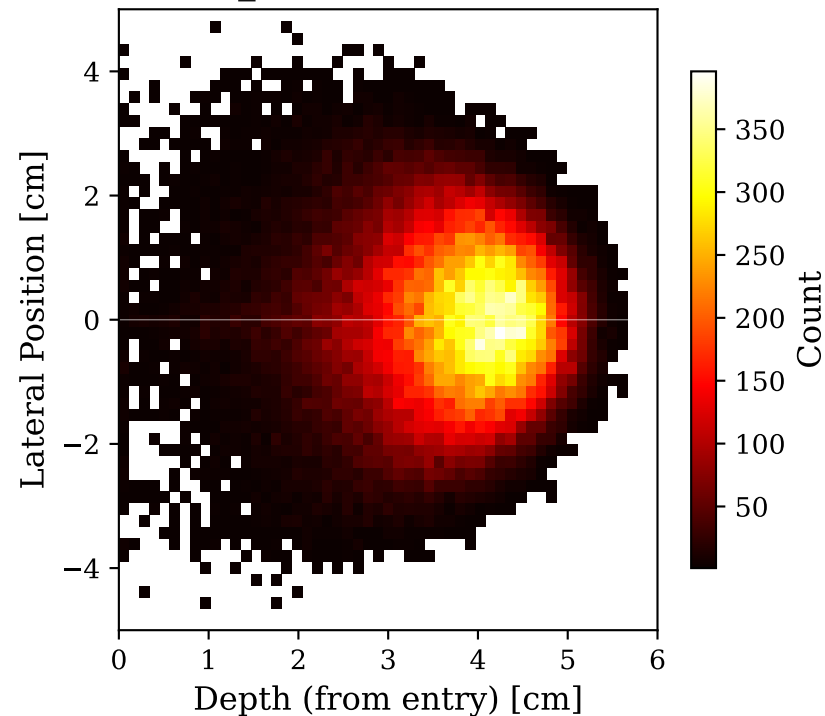
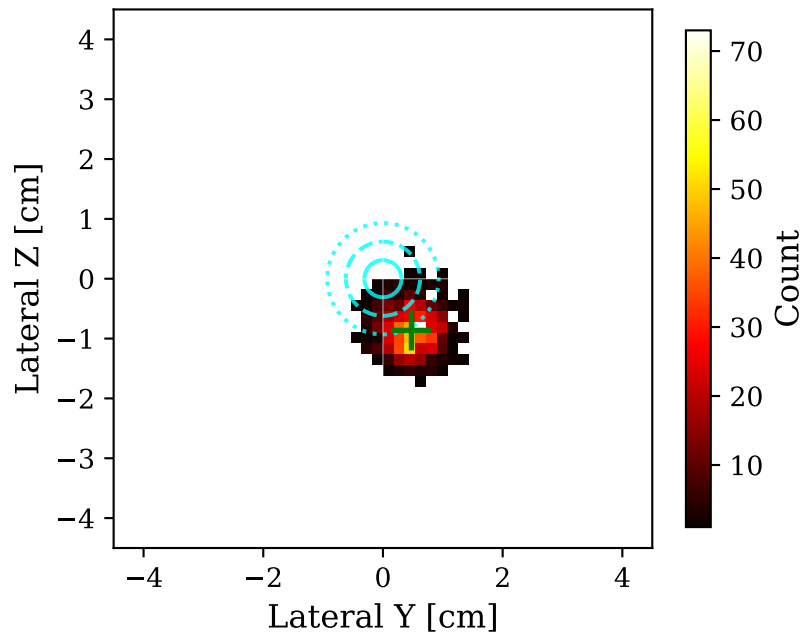


Trajectory Distribution: Geant4 Side View vs ML Agent Cross-Section

Geant4 (Reference)
Side View (like density texture)
 $\sigma_{\text{lateral}} = 1.20 \text{ cm}$



PPO
Cross-Section View
 $\sigma = 0.31 \text{ cm}$
Bias: (+0.48, -0.86)



SAC
Cross-Section View
 $\sigma = 0.25 \text{ cm}$
Bias: (+0.35, +0.76)

